



Graduate elective Introduction to Synergetics

Frequency: weekly, according to the timetable



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Course description. Synergetics studies how macroscopic order, coherent motion, and pattern arise in open nonlinear systems with many interacting degrees of freedom. Following Haken's original route, the laser is the motivating paradigm: above a pump threshold, microscopic atomic polarization and population inversion generate a coherent electromagnetic field; this field becomes an order parameter that in turn organizes the material degrees of freedom. Rayleigh–Bénard convection supplies the canonical spatial counterpart: a maintained heat flux destabilizes conduction and selects organized convection rolls in a viscous, heat-conducting fluid. From these examples the course develops instability, order parameters, the slaving principle, circular causality, normal forms, universality, and the interplay of deterministic dynamics, spatial coupling, and fluctuations. Analytical work is paired with reproducible Python experiments.

Prerequisites. Graduate-level classical and statistical mechanics; ordinary differential equations; linear algebra; multivariable calculus; basic probability. Prior exposure to partial differential equations, stochastic processes, and Python is helpful but not required. The notebooks provide just-in-time theoretical bridges for the required dynamical-systems, Fourier, reaction–diffusion, network, and stochastic-calculus concepts.

Credits: 3 ECTS

Workload: 90 hours = 28 lecture + 14 computational laboratory + 48 independent study

Language of instruction: English

Final control: graded course project and oral defense

Learning outcomes

After completing the course, students will be able to:

- LO1. distinguish control parameters, state variables, and order parameters in open nonlinear systems;
- LO2. determine local stability from linearization and classify codimension-one bifurcations by normal form;

- LO3. derive a reduced amplitude equation by adiabatic elimination or center-manifold reasoning and test its validity numerically;
- LO4. analyze multistability with potential landscapes and elementary catastrophe theory;
- LO5. derive conditions for pattern-forming and diffusion-driven instabilities in spatially extended systems;
- LO6. quantify synchronization, deterministic chaos, fractal structure, and noise-induced switching with appropriate observables;
- LO7. design, document, and defend a reproducible computational investigation of self-organization.

Course organization

Wk.	Lecture focus	Laboratory / independent task	Notebook
1	Pre-Haken pathway: Schrödinger, Turing, and Haken's laser program; contemporary embodied-brain case	Distinguish precursor, mechanism, and demonstration; identify control, microscopic, and order variables	syn_00, syn_01
2	Maxwell–Bloch variables; atomic polarization and inversion; adiabatic elimination; circular causality	Compare full laser dynamics with the near-threshold amplitude equation	syn_01
3	Codimension-one bifurcations; self-oscillations, limit cycles, and the Van der Pol model	Simulate attraction to a limit cycle; measure its period and cycle-averaged energy balance	syn_02
4	Critical slowing down; soft modes; center manifolds; slaving principle	Compare full fast–slow dynamics with a reduced equation	syn_03
5	Potential landscapes; fold and cusp catastrophes; hysteresis	Construct a cusp surface and follow hysteretic paths	syn_04
6	Symmetry, amplitude equations, universality, mode competition	Stuart–Landau saturation and competing modes	syn_05
7	Rayleigh–Bénard convection as an open dissipative structure; spatial instability and wavelength selection; Swift–Hohenberg equation	Visualize the conductive-to-roll bifurcation and simulate a generic finite-wavenumber instability	syn_05
8	Reaction–diffusion systems; Brusselator; Turing instability; morphogenesis	Derive the Brusselator dispersion relation and compare predicted and simulated patterns	syn_06
9	Belousov–Zhabotinsky chemistry; Oregonator oscillations; excitability, fronts, pulses, and refractory dynamics	Simulate a stiff Oregonator and a FitzHugh–Nagumo traveling pulse	syn_07
10	Huygens' clocks and movable-base metronomes; phase reduction, entrainment, and the Kuramoto transition	Explain support-mediated coupling; map pair locking and estimate the collective threshold	syn_08
11	Rayleigh–Bénard convection; Lorenz truncation; strange attractors; Lyapunov exponents	Relate Lorenz variables to convection and compute finite-time divergence	syn_09

Wk.	Lecture focus	Laboratory / independent task	Notebook
12	Fractals, invariant measures, entropy, and time-series diagnostics	Logistic-map bifurcation diagram and dimension estimate	syn_09
13	Langevin and Fokker–Planck descriptions; noise-induced transitions; stochastic resonance	Simulate switching in a noisy bistable potential	syn_10
14	Coupled maps, networks, information processing, and control of chaos	Capstone studio, reproducibility audit, and oral defense	syn_11

Assessment

Component	Points	Primary outcomes
Notebook laboratories (6 submissions)	30	LO1–LO6
Analytical problem sets (2)	20	LO2–LO5
Midterm concept and derivation check	15	LO1–LO5
Capstone report, code, and oral defense	30	LO3–LO7
Research discussion and peer feedback	5	LO1, LO7
Total	100	

Grading scale. A: 90–100; B: 85–89; C: 75–84; D: 70–74; E: 60–69; FX: 35–59; F: 0–34.

Capstone expectations

Students select a physical, chemical, biological, or information-processing system and formulate a synergetic model. The submission must identify a control parameter and an observable order parameter, derive or justify a reduced model, analyze at least one transition, compare theory with numerical evidence, report numerical uncertainty or convergence checks, and state the limits of the model. A fully reproducible notebook and a concise oral defense are required.

Course policies

- **Academic integrity.** Students must follow the university’s [Code of Academic Integrity](#). Sources, reused code, and AI assistance must be disclosed. Submitted analysis must be understood and defensible by the student.
- **Reproducibility.** Computational work must run from a clean kernel, use fixed seeds where randomness is not itself under study, label axes and units, and explain parameter choices.
- **Attendance and deadlines.** Students are responsible for missed material. Extensions should be arranged before the deadline when possible; documented university policies take precedence.
- **Communication.** Official university email and the designated learning-management space are the authoritative channels for materials and announcements.

Principal literature

1. E. Schrödinger, *What Is Life? The Physical Aspect of the Living Cell*, Cambridge University Press (1944; reissued 2012), [publisher page](#).
2. H. Haken, *Synergetics: An Introduction*, 3rd ed., Springer Series in Synergetics 1 (1983), [doi:10.1007/978-3-642-88338-5](#).
3. H. Haken, *Synergetics: Introduction and Advanced Topics*, Springer (2004), [doi:10.1007/978-3-662-10184-1](#).
4. A. S. Mikhailov, *Foundations of Synergetics I: Distributed Active Systems*, 2nd ed., Springer (1990), [doi:10.1007/978-3-642-78556-6](#).
5. A. S. Mikhailov and A. Yu. Loskutov, *Foundations of Synergetics II: Complex Patterns*, Springer (1991).
6. R. Gilmore, *Catastrophe Theory for Scientists and Engineers*, Wiley (1981); Dover reprint (1993), ISBN 978-0-486-67539-8.
7. M. C. Cross and P. C. Hohenberg, “Pattern formation outside of equilibrium,” *Rev. Mod. Phys.* **65**, 851 (1993), [doi:10.1103/RevModPhys.65.851](#).
8. B. Wen, D. Goluskin, and C. R. Doering, “Steady Rayleigh–Bénard convection between no-slip boundaries,” *J. Fluid Mech.* **933**, R4 (2022), [doi:10.1017/jfm.2021.1042](#).
9. Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence*, Springer (1984), [doi:10.1007/978-3-642-69689-3](#).
10. J. Pantaleone, “Synchronization of metronomes,” *Am. J. Phys.* **70**, 992–1000 (2002), [DOI](#).
11. A. M. Turing, “The Chemical Basis of Morphogenesis,” *Phil. Trans. R. Soc. B* **237**, 37 (1952), [doi:10.1098/rstb.1952.0012](#).
12. I. Prigogine and R. Lefever, “Symmetry Breaking Instabilities in Dissipative Systems. II,” *J. Chem. Phys.* **48**, 1695–1700 (1968), [doi:10.1063/1.1668896](#).
13. A. M. Zhabotinsky, “Belousov–Zhabotinsky reaction,” *Scholarpedia* **2**(9), 1435 (2007), [DOI](#).
14. E. N. Lorenz, “Deterministic Nonperiodic Flow,” *J. Atmos. Sci.* **20**, 130–141 (1963), [DOI](#).
15. C. Gardiner, *Stochastic Methods: A Handbook for the Natural and Social Sciences*, 4th ed., Springer (2009), ISBN 978-3-540-70712-7.
16. H. Risken, *The Fokker–Planck Equation: Methods of Solution and Applications*, 2nd ed., Springer (1996), [doi:10.1007/978-3-642-61544-3](#).
17. N. G. van Kampen, *Stochastic Processes in Physics and Chemistry*, 3rd ed., Elsevier (2007), ISBN 978-0-444-52965-7.
18. M. Breakspear, “Dynamic models of large-scale brain activity,” *Nature Neuroscience* **20**, 340–352 (2017), [doi:10.1038/nn.4497](#).
19. A. Arenas, A. Díaz-Guilera, J. Kurths, Y. Moreno, and C. Zhou, “Synchronization in complex networks,” *Physics Reports* **469**, 93–153 (2008), [doi:10.1016/j.physrep.2008.09.002](#).

20. D. Helbing, “Traffic and related self-driven many-particle systems,” *Rev. Mod. Phys.* **73**, 1067–1141 (2001), [doi:10.1103/RevModPhys.73.1067](https://doi.org/10.1103/RevModPhys.73.1067).
21. MIT OpenCourseWare, [Nonlinear Dynamics I: Chaos](#), supplementary notes and problems.



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Course teacher

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